

Mathematics for Cryptography

Exam Notes: Unit 1

Modular Arithmetic · Euclid · Extended Euclid · Fermat · Euler · CRT · RSA link

Kaise padhein: Har topic mein pehle **Definition**, phir **Hinglish samajh**, phir **Formula**, phir **Solved example** aur end mein **Exam tips** hain. Last page par ek **quick revision sheet** hai.

#	Topic
1	Modular Arithmetic
2	Euclidean Algorithm (GCD)
3	Extended Euclidean Algorithm
4	Fermat's Little Theorem
5	Euler's Totient & Euler's Theorem
6	Chinese Remainder Theorem
7	RSA aur in sabka role
8	Quick Revision Sheet & Practice

1. Modular Arithmetic

Definition (English): Integers a and b are **congruent modulo n** , written $a \equiv b \pmod{n}$, if n divides $(a - b)$. $a \bmod n$ is the remainder when a is divided by n , in the range 0 to $n-1$.

Hinglish samajh: Mod ka matlab: number ko n se divide karo aur **sirf remainder** dekho. Ghadi (clock) mein 13:00 ko 1 baje bolte hain, kyunki 12 ke baad wapas shuru. Yani $13 \bmod 12 = 1$.

Formulas

- $(a + b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$
- $(a \times b) \bmod n = [(a \bmod n) \times (b \bmod n)] \bmod n$
- $a^k \bmod n = (a \bmod n)^k \bmod n$
- Inverse:** $a^{-1} \bmod n = x$ such that $a \cdot x \equiv 1 \pmod{n}$. Exists **only if $\gcd(a, n) = 1$** .

Example: $23 \times 17 \bmod 7$

- $23 \bmod 7 = 2$
- $17 \bmod 7 = 3$
- $2 \times 3 = 6 \rightarrow \text{Answer} = 6$

Tip: Hamesha pehle numbers ko mod se chhota karo, phir multiply karo.

2. Euclidean Algorithm (GCD)

Definition (English): A method to find $\gcd(a, b)$ using the rule **$\gcd(a, b) = \gcd(b, a \bmod b)$** , repeated until the remainder is 0. The last non-zero remainder is the GCD.

Hinglish samajh: Bade number ko chhote se divide karo, phir chhote ko remainder se divide karo. Yeh tab tak karo jab remainder 0 na ho. **Aakhri non-zero remainder = GCD.**

Example: $\gcd(270, 192)$

Step	Division	Remainder
1	$270 = 1 \times 192 + 78$	78
2	$192 = 2 \times 78 + 36$	36
3	$78 = 2 \times 36 + 6$	6
4	$36 = 6 \times 6 + 0$	0

Last non-zero remainder = 6, so **$\gcd(270, 192) = 6$** .

3. Extended Euclidean Algorithm

Definition (English): Finds $\gcd(a, b)$ and integers x, y such that **$ax + by = \gcd(a, b)$** (Bézout's identity). Main use: computing **modular inverses**, e.g. RSA private key.

Hinglish samajh: Euclid sirf GCD batata hai. Extended Euclid batata hai ki GCD ko a aur b ke combination se kaise likhein. Agar $\gcd = 1$ aaye, to $ax + by = 1$ ka matlab **x hi a ka inverse hai $(\bmod b)$** .

Example: Inverse of 7 mod 40

Step 1 (Euclid): $40 = 5 \times 7 + 5$; $7 = 1 \times 5 + 2$; $5 = 2 \times 2 + 1$; $2 = 2 \times 1 + 0$. $\gcd = 1$, to inverse exists.

Step 2 (back-substitute, 1 se shuru):

- $1 = 5 - 2 \times 2$
- $2 = 7 - 5$ daalo: $1 = 5 - 2(7 - 5) = 3 \times 5 - 2 \times 7$
- $5 = 40 - 5 \times 7$ daalo: $1 = 3(40 - 5 \times 7) - 2 \times 7 = \mathbf{3 \times 40 - 17 \times 7}$

Step 3: $7 \times (-17) \equiv 1 \pmod{40}$. $-17 + 40 = 23$, so **inverse = 23**. Check: $7 \times 23 = 161 = 4 \times 40 + 1$ ✓

4. Fermat's Little Theorem

Definition (English): If **p** is prime and $\gcd(a, p) = 1$, then $a^{p-1} \equiv 1 \pmod{p}$. Equivalent: $a^p \equiv a \pmod{p}$ for any a .

Hinglish samajh: Agar p prime hai aur a, p ka multiple nahi, to a ko $(p-1)$ baar multiply karke p se divide karo, remainder hamesha **1**. Powers ek **cycle** mein chalte hain jiski length $p-1$ hai. Jaise $p=7$: 2 ki powers ka remainder 2, 4, 1, 2, 4, 1... aur $2^6=64 = 9 \times 7 + 1$.

Formulas

- **Big power:** $a^n \bmod p = a^{(n \bmod (p-1))} \bmod p$
- **Inverse:** $a^{-1} \equiv a^{p-2} \pmod{p}$
- **Primality test:** if $a^{n-1} \not\equiv 1 \pmod{n}$, then n is composite

4-step method

- $p-1$ nikalo
- Power ko $(p-1)$ se divide karo, **remainder r** lo
- Remainder 0 ho to answer **1**; warna a^r calculate karo
- p se mod karo

Example 1: $2^{100} \bmod 7$

- $p-1 = 6$; $100 = 16 \times 6 + 4$
- $2^{100} \equiv 2^4 = 16$
- $16 \bmod 7 = 2$

Example 2: $3^{50} \bmod 7$

- $50 = 8 \times 6 + 2$
- $3^2 = 9$; $9 \bmod 7 = 2$

Example 3: Inverse of 4 mod 11

- $4^9 \bmod 11$: $4^2=16 \equiv 5$, $4^4 \equiv 25 \equiv 3$, $4^8 \equiv 9$, $4^9 \equiv 9 \times 4 = 36 \equiv 3$
- **Inverse = 3**. Check: $4 \times 3 = 12 \equiv 1 \checkmark$

Galtiyan: (1) p prime hona chahiye. (2) Exponent **(p-1)** se reduce hota hai, p se nahi. (3) Agar $p \mid a$ to answer 0 hai, 1 nahi. (4) Converse false hai: Carmichael numbers (jaise 561) Fermat test ko fool karte hain.

5. Euler's Totient and Euler's Theorem

Definition (English): $\phi(n)$ = count of integers in $1..n$ that are coprime to n . **Euler's Theorem:** if $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$.

Hinglish samajh: $\phi(n) = 1$ se n tak ke woh numbers ki ginti jinka n ke saath koi common factor nahi. Euler ka theorem Fermat jaisa hi hai, bas modulus prime hona zaruri nahi aur **cycle length p-1 ki jagah $\phi(n)$** hai. Fermat, Euler ka special case hai.

$\phi(n)$ ke rules

Case	Formula	Example
$n = p$ (prime)	$\phi(p) = p - 1$	$\phi(7) = 6$

Case	Formula	Example
$n = p^k$	$p^k - p^{k-1}$	$\phi(9) = 9 - 3 = 6$; $\phi(8) = 8 - 4 = 4$
$n = m \cdot k, \gcd(m, k) = 1$	$\phi(m) \cdot \phi(k)$	$\phi(12) = \phi(4) \cdot \phi(3) = 2 \cdot 2 = 4$
$n = p \cdot q$ (RSA)	$(p-1)(q-1)$	$\phi(15) = 2 \cdot 4 = 8$

Counting method: $\phi(10)$

1 se 10 tak, jo 10 ke saath common factor rakhte hain unhe kaato. Bache: 1, 3, 7, 9. Total 4, so **$\phi(10) = 4$** .

4-step method (numericals ke liye)

- $\gcd(a, n) = 1$ check karo
- $\phi(n)$ nikalo
- Power ko $\phi(n)$ se divide karo, remainder r lo ($r = 0$ ho to answer 1)
- $a^r \bmod n$

Example 1: $7^{222} \bmod 10$

- $\phi(10) = 4$; $222 = 55 \times 4 + 2$
- $7^2 = 49$; $49 \bmod 10 = \mathbf{9}$

Example 2: $2^{100} \bmod 9$

- $\phi(9) = 6$; $100 = 16 \times 6 + 4$
- $2^4 = 16$; $16 \bmod 9 = \mathbf{7}$
- (2 ki powers mod 9: 2, 4, 8, 7, 5, 1 repeat)

Example 3: $5^{50} \bmod 12$

- $\phi(12) = 4$; $50 = 12 \times 4 + 2$
- $5^2 = 25$; $25 \bmod 12 = \mathbf{1}$

	Fermat	Euler
Modulus	Sirf prime p	Koi bhi n
Cycle length	$p - 1$	$\phi(n)$
Formula	$a^{p-1} \equiv 1$	$a^{\phi(n)} \equiv 1$
Shart	$\gcd(a, p) = 1$	$\gcd(a, n) = 1$

6. Chinese Remainder Theorem (CRT)

Definition (English): Let $m_1, m_2, \dots, m_k$ be **pairwise coprime**. Then the system $x \equiv a_1 \pmod{m_1}, \dots, x \equiv a_k \pmod{m_k}$ has a **unique solution modulo $M = m_1 \cdot m_2 \cdot \dots \cdot m_k$** .

Hinglish samajh: Agar alag-alag chhote moduli se divide karne par remainders pata hain (moduli coprime hon), to woh **original number ek hi milta hai** (M ke andar). Do chhoti conditions ko milakar ek badi condition banana.

Formula

- $M = m_1 \times m_2 \times \dots \times m_k$
- $M_i = M / m_i$
- $y_i = M_i^{-1} \pmod{m_i}$ (Extended Euclid ya trial se)
- $x = (a_1 M_1 y_1 + a_2 M_2 y_2 + \dots) \pmod{M}$

Example 1: $x \equiv 2 \pmod{3}, x \equiv 3 \pmod{5}$

- $M = 15; M_1 = 5, M_2 = 3$
- $y_1: 5y \equiv 1 \pmod{3} \rightarrow 2y \equiv 1 \rightarrow y_1 = 2$
- $y_2: 3y \equiv 1 \pmod{5} \rightarrow y_2 = 2$
- $x = 2 \cdot 5 \cdot 2 + 3 \cdot 3 \cdot 2 = 20 + 18 = 38$
- $38 \pmod{15} = 8$. Check: $8 \pmod{3} = 2 \checkmark, 8 \pmod{5} = 3 \checkmark$

Example 2: $x \equiv 1 \pmod{3}, x \equiv 4 \pmod{5}, x \equiv 2 \pmod{7}$

- $M = 105; M_1 = 35, M_2 = 21, M_3 = 15$
- $y_1: 35y \equiv 1 \pmod{3} \rightarrow 2y \equiv 1 \rightarrow y_1 = 2; y_2: 21y \equiv 1 \pmod{5} \rightarrow y_2 = 1; y_3: 15y \equiv 1 \pmod{7} \rightarrow y_3 = 1$
- $x = 1 \cdot 35 \cdot 2 + 4 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 70 + 84 + 30 = 184$
- $184 \pmod{105} = 79$. Check: $79 \pmod{3} = 1 \checkmark, \pmod{5} = 4 \checkmark, \pmod{7} = 2 \checkmark$

Exam format: (1) moduli coprime check $\rightarrow$ (2) $M \rightarrow$ (3) $M_i \rightarrow$ (4) $y_i \rightarrow$ (5) $x = \sum a_i M_i y_i \pmod{M} \rightarrow$ (6) Verify.

7. RSA aur in sabka role

Definition (English): RSA is a public-key cryptosystem. Public key (e, n) encrypts; private key (d, n) decrypts. Security rests on the difficulty of **factoring $n = p \cdot q$** .

Hinglish samajh: Taala-chaabi jaisa: open taala (public key) sabko de do, koi bhi box band kar sakta hai. Kholne ki chaabi (private key) sirf tumhare paas. $n = p \times q$ se wapas p, q nikalna bahut mushkil hai.

RSA steps

- Do primes p, q chuno; $n = p \cdot q$
- $\phi(n) = (p-1)(q-1)$
- e chuno: $1 < e < \phi(n), \gcd(e, \phi(n)) = 1$
- $d = e^{-1} \pmod{\phi(n)}$
- Encrypt: $C = M^e \pmod{n}$ Decrypt: $M = C^d \pmod{n}$

Example: $p=3, q=11$

- $n = 33, \phi = 20; e = 3; d = 7$ ($3 \times 7 = 21 \equiv 1 \pmod{20}$)

- $M = 4$: $C = 4^3 \bmod 33 = 64 \bmod 33 = \mathbf{31}$
- Decrypt: $31^7 \bmod 33 = \mathbf{4}$ ✓

Kaun sa tool kahan kaam aata hai

Tool	RSA mein kaam
Modular arithmetic	Poora RSA isi mein hota hai
Euclid (GCD)	$\gcd(e, \phi(n)) = 1$ check karna
Extended Euclid	Private key d nikalna
Fermat	Prime testing (Fermat / Miller-Rabin); CRT decryption mein exponent d ko $d \bmod (p-1)$ karna
Euler	RSA correct kyun hai (proof)
CRT	Decryption ~ 4 guna fast karna ($\bmod p$ aur $\bmod q$ alag, phir jodna)

Correctness proof (5 marks)

- $e \cdot d \equiv 1 \pmod{\phi(n)}$, so $ed = 1 + k\phi(n)$
- $C^d = M^{ed} = M \cdot (M^{\phi(n)})^k$
- Euler: $M^{\phi(n)} \equiv 1 \pmod{n}$
- So $C^d \equiv M \cdot 1^k = \mathbf{M} \pmod{n}$ ■

8. Quick Revision Sheet

Topic	Key idea / formula	Use
Modular	$a \equiv b \pmod{n}$ if $n \mid (a-b)$	Foundation
Euclid	$\gcd(a,b) = \gcd(b, a \bmod b)$	GCD
Ext. Euclid	$ax + by = \gcd(a,b)$	Inverse, RSA d
Fermat	$a^{p-1} \equiv 1 \pmod{p}$; exponent mod $(p-1)$	Fast powers, prime test
Euler	$a^{\phi(n)} \equiv 1 \pmod{n}$; exponent mod $\phi(n)$	RSA correctness
$\phi(n)$	$\phi(p)=p-1$; $\phi(pq)=(p-1)(q-1)$; $\phi(p^k)=p^k - p^{k-1}$	Euler, RSA
CRT	$x = \sum a_i M_i y_i \bmod M$	Solve systems, fast RSA

Exam Tips

- Hamesha pehle mod se numbers chhote karo, phir multiply/power karo.
- Inverse tabhi exist karta hai jab **gcd = 1**. Pehle check karo.
- Modulus prime $\rightarrow$ Fermat. Modulus composite $\rightarrow$ Euler.
- Exponent **(p-1)** ya **$\phi(n)$** se reduce hota hai, modulus se nahi.
- Exponent ka remainder 0 ho to answer **1**.
- CRT mein answer ko har original congruence par verify karo.
- Marks step-wise milte hain: har step likho.

Practice Questions (answers neeche)

- Find $3^{17} \bmod 5$.
- Find $\phi(12)$, $\phi(15)$, $\phi(25)$.
- Find $3^{100} \bmod 10$.
- Find inverse of 5 mod 17 (Extended Euclid).
- Solve $x \equiv 1 \pmod{2}$, $x \equiv 2 \pmod{3}$.
- RSA: $p=5$, $q=11$, $e=3$. Find n , $\phi(n)$, d . Encrypt $M=9$.

Answers

- 3^{17} : $17 = 4 \times 4 + 1 \rightarrow 3^1 = 3 \rightarrow \mathbf{3}$
- $\phi(12)=\mathbf{4}$, $\phi(15)=\mathbf{8}$, $\phi(25)=\mathbf{20}$
- $\phi(10)=4$; $100 = 25 \times 4 + 0 \rightarrow \mathbf{1}$
- $5 \cdot 7 = 35 = 2 \times 17 + 1 \rightarrow \text{inverse} = \mathbf{7}$
- $x = \mathbf{5}$ ($5 \bmod 2 = 1$, $5 \bmod 3 = 2$), so $x \equiv 5 \pmod{6}$
- $n = \mathbf{55}$, $\phi = 4 \times 10 = \mathbf{40}$, d : $3d \equiv 1 \pmod{40} \rightarrow d = \mathbf{27}$. $C = 9^3 \bmod 55 = 729 \bmod 55 = 729 - 715 = \mathbf{14}$